

Félix LAPLANTE

PERSONAL INFORMATION

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EDUCATION

2024–2026	Master in Mathematics and AI, Orsay Joint coursework with the Probability and Statistics program, co-organized with Institut Polytechnique de Paris.	Ranked 1 st , 19.28/20 in M1, 18.28 in M2
2021–2024	Bachelor in Mathematics and Applications, Évry	Ranked 1 st , 19.425/20 in L3

RESEARCH EXPERIENCE

APRIL–SEPTEMBER 2026	Research internship in Causality under Christophe Ambroise and Pierre Humbert, LaMME Study of identifiability in structural causal models, with a focus on non-Gaussian settings and causal discovery. Analysis of quantum causal inflation constraints and their implications for classical causal representations.
APRIL–SEPTEMBER 2025	Research internship on Joint Models under Estelle Kuhn, Christophe Ambroise, INRAE Development of estimation and prediction methods for joint models and extension to multi-state models. Derivation of uniform high-probability bounds for survival functions, implementation of the <code>jmsstate</code> package, simulation studies and analysis on real-world data, and comparison with state-of-the-art methods.
APRIL–JUNE 2024	Research internship in Clustering under Christophe Ambroise, LaMME Design and implementation of a hierarchical topological clustering algorithm inspired by SVMs, <i>Spectral Bridges</i> . Partitioning into Voronoi regions, followed by grouping and automatic detection of boundaries and low-density regions using spectral methods on a local scale-invariant affinity matrix.

AWARDS / GRANTS

JUNE 2024 | Sophie Germain Excellence Scholarship

PUBLISHED PAPERS / PREPRINTS

MAY 2026	A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores <i>Félix Laplante</i> arXiv preprint https://arxiv.org/abs/2605.25852 .
OCTOBER 2025	A General Framework for Joint Multi-State Models <i>Félix Laplante, Christophe Ambroise</i> arXiv preprint https://arxiv.org/pdf/2510.07128 submitted to <i>Statistics and Computing</i> .
DECEMBER 2024	Scalable Spectral Clustering Based on Vector Quantization <i>Félix Laplante, Christophe Ambroise</i> <i>Computo</i> journal paper https://doi.org/10.57750/1gr8-bk61 .

TALKS / POSTERS

JUNE 2026 | A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores

JUNE 2026	Poster presentation at Ukrainian Mathematics Days (IHP) discussing the PIT-CP framework for conditional validity based on pivotal scores.
	A General Framework for Joint Multi-State Models
SEPTEMBER 2025	Oral presentation at <i>Journées de la Statistique</i> (Clermont-Ferrand) introducing a unified framework for longitudinal biomarkers and multi-state event processes.
	Uniform High-Probability Bands for Survival Functions
JUNE 2025	Poster presentation at <i>StatMathAppli</i> (Fréjus) deriving asymptotic uniform confidence bands under a joint modeling framework.
	Selecting the Number of Clusters in a Clustering Algorithm
	Oral presentation at <i>Journées de la Statistique</i> (Marseille) introducing a connectivity-based score for scalable spectral clustering.

OTHER ATTENDED EVENTS

MARCH 2026	Workshop on Uncertainty Quantification and Causality
OCTOBER 2025	Winter School on Causality and Explainable AI
JULY 2025	Working group on Topological Data Analysis (LaMME)
JUNE 2025	Working group on Causality (MIA Paris-Saclay)

SERVICES

MAY 2026	Co-organizer of a Causality Working Group with Pierre Humbert
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PACKAGES

PYTHON	jmstate Companion package to the <i>Multi-State Models</i> paper, built on PyTorch. pitcp Companion package to the <i>Pivotal Scores Conformal Prediction</i> paper, built on zuko. sbcluster Companion package to the <i>Scalable Spectral Clustering</i> paper. fastkmeanspp C++-backed implementation of K-means++, with massive speedups over <code>scikit-learn</code>. mimisbm Vectorized port of the corresponding R project, implementing a mixture of multilayer stochastic block models. uniformbands Asymptotic uniform confidence bands for survival functions.
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LANGUAGES

FRENCH	Native
ENGLISH	Native

INTERESTS & ACTIVITIES

HOBBIES	Photography, cycling, piano, hiking
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